JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of CSE
"A Place to Learn; A Chance to Grow"
2024-25 (ODD)

VISION	MISSION
" To be recognized as an Excellent Innovative Engineering Department through Academic Programme which develops leaders in Education and Research in Computer Science "	1. To create self-learning environment by facilitating leadership qualities, team-spirit and ethical responsibilities. 2. To strengthen department-industry collaboration and interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Question Bank

Semester: V Semester **Branch: CSE**

Subject: DAA **Subject Code: CS5T003**

Q. No.	Questions
	UNIT I
1	Define algorithm? List out the different design techniques? (2 M)
2	Define worst case, average case and best case analysis? (2 M)
3	Define growth rate of function. (2 M)
4	Differentiate between algorithm and program. (2 M)
	Differentiate between recursive and iterative algorithm design. (2 M)
5	List and explain different types of complexities used to find efficiency of algorithm. (2 M)
6	Define Asymptotic Notation. (2 M)
7	Explain the characteristics of an algorithm. (2 M)
8	Differentiate Time Efficiency and Space Efficiency. (2 M)
9	Explain recursion with example. (2 M)
10	Define Amortized Analysis. List out the method of Amortized Analysis.
11	Discuss the need of Analysing Algorithms.
12	Define Asymptotic Notations. Explain different types of Asymptotic Notations.(4M)
13	Explain the characteristics of an algorithm. (4 M)
14	"Constants are ignored in analysis of algorithm during complexity Computation." Justify with example. (4 M)
15	Define the term Recursion. (2 M)
16	Define algorithm? How to measure algorithm's running time?(4 M)

17	Solve the following equation by using characteristic equation. (7 M) $t_n = n$, if $n = 0$ or $n = 0$ or $n = 2$ $= 5t_{n-1} - 8t_{n-2} + 4t_{n-3}$, otherwise.
18	List and explain different types of complexities used to find efficiency of algorithm.(4 M)
19	Discuss the need of Analyzing Algorithms. (4 M)
20	Explain the Amortized Analysis & different methods used in Amortized analysis.(7 M)
21	Solve the following recurrence equation: $T(n) = T(n/4) + \sqrt{n} + 1 \quad \text{if} \quad n \geq 4$ Subject to initial condition T(n)= 1 for n= 1 Use technique of characteristic equation.
UNIT II	
1	Describe the steps of Divide and Conquer Strategy.(2 M)
2	Write the complexity of a) Selection sort b) Merge sort.(2 M)
3	Define External Path Length and Internal Path Length. .(2 M)
4	Explain the steps of sorting array using Merge Sort. .(2 M)
5	Apply Quick Sort on the following array. (4 M) 40, 20, 10, 80, 60, 50, 7, 30, 100.
6	Write an algorithm for Binary Search. (4 M)
7	Draw split & merge tree for array of 10 elements. (4 M)
8	Apply merge sort on the following array (4 M) 100, 300, 150, 450, 250, 200, 500, 400.
9	Find the average number of successful and unsuccessful search. Also find internal & external path length. Consider the array given below to perform binary search: a = [-15, -6, 0, 7, 9, 23, 54, 82, 101, 112, 125, 131, 142, 151] Also give the complexity of binary search. (7 M)
10	Find out average number of comparisons required for successful and unsuccessful binary search on following array. -12, 23, 31, 45, 56, 78, 90, 103, 113, 126, 157(7 M)
11	Explain Analysis of binary search for following elements and calculate Avg. no. of successful and Avg no. of unsuccessful companisions -10, 25, 15, 16, 18, -9, 4, 2. (7 M)
12	Explain the Divide and Conquer Strategy in short. (4 M)
13	Write the complexity of a) Selection sort b) Merge sort (2 M)
14	Apply merge sort on the following array. (4 M) 23, 65, 97, 58, 39, 13, 89, 43, 16, 73
15	Find the average number of successful and unsuccessful search. Also find internal & external path length. Consider the array given below to perform binary search: (12 M) A = [15 6 0 7 9 23 54 82 101 112 125 131 142 151] Also give the complexity of binary search
16	Differentiate between Divide & Conquer and Dynamic Programming (4 M)
UNIT III	
1	List the advantage of the greedy algorithm. (2 M)

2	List out the methods for solving Knapsack Problem. (2 M)
3	Write the principle of Travelling Salesman Problem. (2 M)
4	Write the difference between a breadth-first search and a depth-first search. (2 M)
5	Explain Spanning tree and Minimum Cost Spanning Tree with example. (2 M)
6	Define Minimum Cost Spanning Tree with example. (2 M)
7	List out the three methods of solving Knapsack Problem. (2 M)
8	Find the optimal solution using greedy algorithm. (4 M) Let n = 4, profit pi [10, 33, 11, 40] and deadlines di [3, 1, 2, 2] respectively.
9	Discuss the characteristics of Greedy Strategy. (4 M)
10	Explain Huffman coding for the data given below. (4 M) a : 26, b : 42, c :84, d :12, e :19, f : 35
11	Explain the Knapsack algorithm to find an optimal solution. Find optimal solution to the knapsack instance. No. of objects = 7, capacity = 15

Objects	1	2	3	4	5	6	7
Weight	2	3	5	7	1	4	1
Profit	10	5	15	7	6	18	3

(7 M)

12 Schedule the following jobs using job scheduling algorithm. Also write the algorithm for Job Scheduling.

Job	Profit	Dead line
1	10	2
2	50	1
3	9	2
4	26	3
5	18	2

(7 M)

13 Differentiate between following. (7 M)

i) Greedy & dynamic programming.

ii) Homogeneous & nonhomogeneous recurrences.

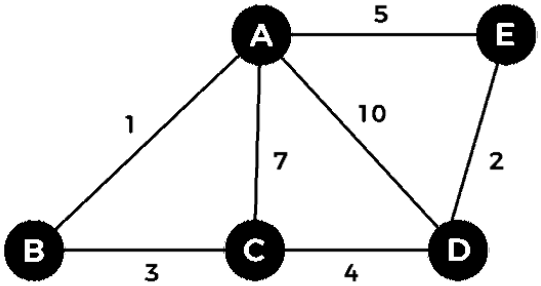
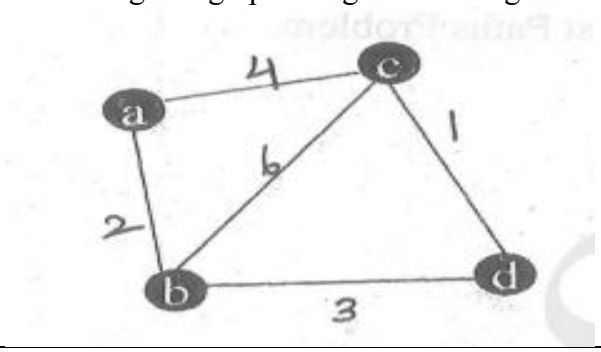
iii) Divide and conquer & dynamic programming.

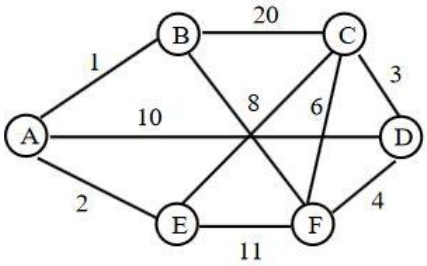
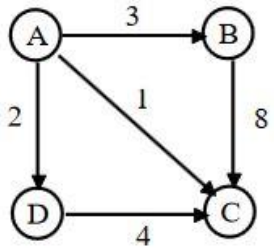
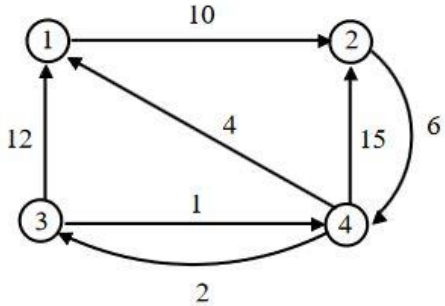
14 Find the multiplication of the following matrices A & B given below using Stressen's matrix multiplication Algorithm.

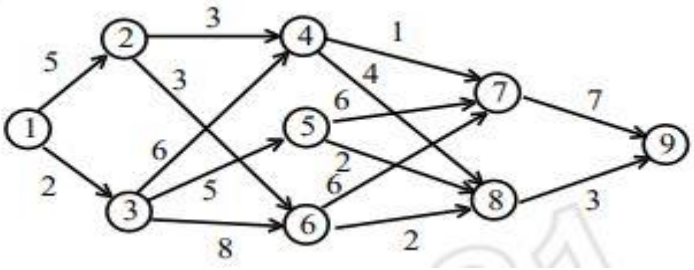
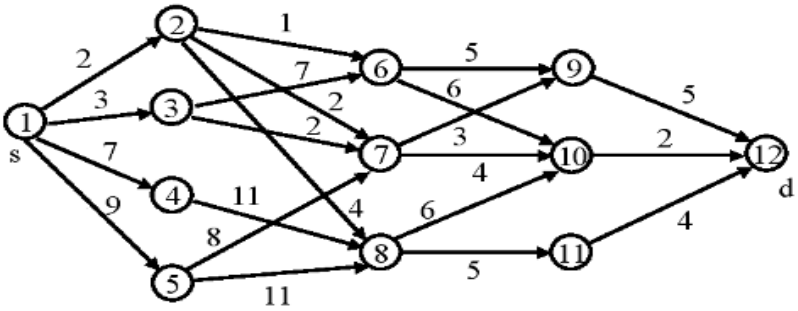
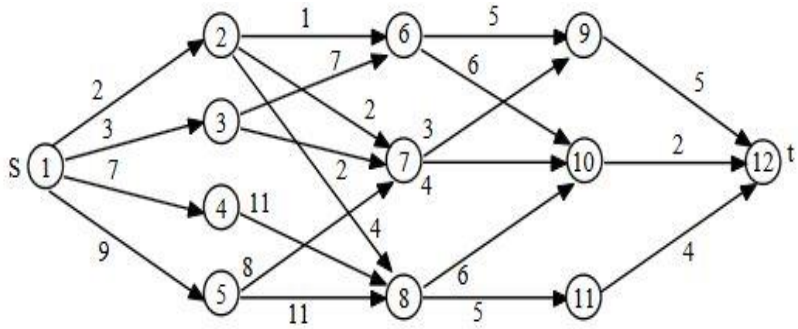
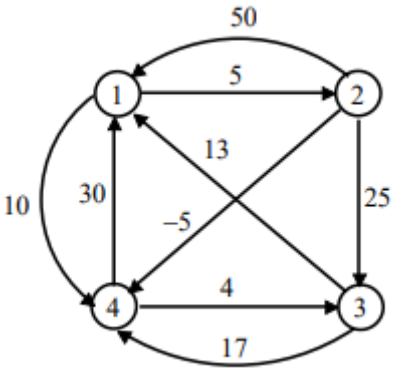
$$A = \begin{bmatrix} 4 & 5 \\ 7 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 \\ 5 & 9 \end{bmatrix}$$

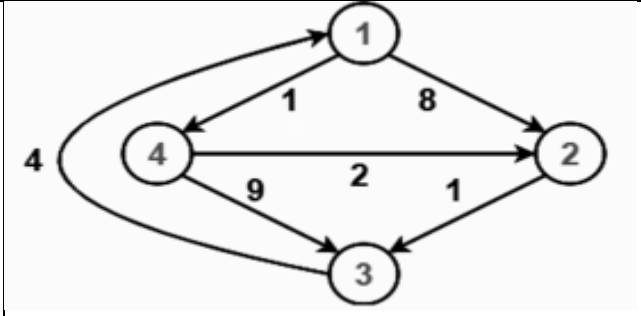
(7 M)

15 List methods to find MST. Find the minimum spanning tree for a given graph using Kruskal's method? (7 M)

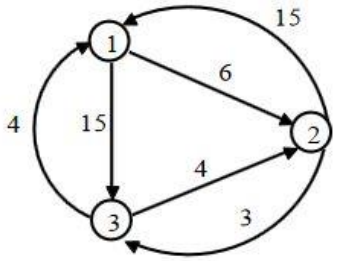
	
16	Explain job scheduling (job sequencing) with deadline problem in short. (4 M) Let n = 4, profit pi [10, 33, 11, 40] and deadlines di [3, 1, 2, 2] respectively. Find the optimal solution using greedy algorithm.
17	Explain the Knapsack algorithm to find an optimal solution. Solve the fractional knapsack problem. Given a knapsack with max weight capacity M = 5 and data contains the items along with their value and weight. M = 5, n = 4, Value = {100, 20, 60, 40} and Weight = {3, 2, 4, 1}. (7 M)
18	Define Greedy Algorithm. (2 M)
19	State the Principle of Job Sequencing with Deadline. (2 M)
20	Define Version Space. (2 M)
21	Define minimum cost spanning trees? List methods to find MST. Find the minimum spanning tree for a given graph using Prism's Algorithm? (4 M) 
22	Define Minimum Cost Spanning Tree with example.(2 M)
23	Find an optimal Huffman code for the following set of frequency. a 50, b: 20, c: 15, d: 30. (4 M)
24	Implement Travelling Salesman Problem on following matrix. (7 M) $\begin{bmatrix} 0 & 8 & 16 & 15 \\ 14 & 0 & 9 & 12 \\ 7 & 10 & 0 & 6 \\ 11 & 13 & 10 & 0 \end{bmatrix}$
25	Explain Minimum Cost Spanning tree. Find the minimum cost spanning tree using i) Kruskal algorithm ii) Prims algorithm for the graph given below : (8 M)

	
26	Explain Huffman coding for the data given below. (5 M) a : 26, b : 42, c :84, d :12, e :19, f : 35 Use variable length encoding method
27	Explain the different types of Graph traversal techniques. (4 M)
Unit IV	
1	Write the principle of Dynamic Programming. (2 M)
2	Write the principle of Travelling Salesman Problem. (2 M)
3	State the principle of Backward and Forward Multistage Graph. (2 M)
4	Write difference between top-down and bottom-up dynamic programming approaches. (2 M)
5	Differentiate between Dynamic Programming and Divide & Conquer Strategy. (4 M)
6	Define Longest Common Subsequence? Find the LCS length for given strings. (7 M) String A = EXPONENTIAL String B = POLYNOMIAL
7	Write the principle of APSP. Implement Floyd – Warshall algorithm on for the following graph. (7 M) 
8	Write an algorithm for Floyd Warshall & Solve following. (7 M) 
9	Determine LCS of X = EXPONENTIAL Y = POLINOMIAL Write recurrence relation for LCS.
10	Find path from source to destination by applying Backward Algorithm on following graph.

	
11	Define multistage graph in dynamic programming? Solve the following graph problem using forward and backward approach. 
12	For the following graph, find the shortest path from source 's' to destination 't' using forward and backward multistage graph algorithm. Write its algorithm & also discuss its complexity.(12 M) 
13	Define Longest Common Subsequence? Find the LCS length for given strings. String X = b, q, d, r, c, v, e, f, g, h String Y = a, b, c, v, d, e, f, g, h
14	Write an algorithm for All Pairs shortest Path and generate a final matrix for the following graph. 
15	Find all pairs shortest path using Floyd Warshall algorithm for given directed graph?



16 Write an algorithm for Floyd Warshall and calculate distance and path matrix of following.(7M)



17 For the following knapsack sequence of objects find profit by three methods. (7 M)
Capacity = 11, n = 4 w = (1, 2, 5, 8, 7) p = (1, 6, 18, 25, 30)

18 Write an algorithm to generate LCS. Apply algorithm for following string & generate LCS.
X = a, a, b, a, a, b, a, b, a, a.
Y = b, a, b, a, a, b, a, b.

19 Find the longest common subsequence for input sequences.
sequence 1 - A G G T A B Z
sequence 2 - G X T X A Y B
Also state the length of longest common subsequence.

20 A knapsack has capacity m = 18, no. of objects = 7. The weight and profit associated is

Object	1	2	3	4	5	6	7
Weight	2	3	5	4	3	6	3
Profit	9	15	12	4	6	16	8

Calculate the total profit using following approaches -
i) Maximum Profit
ii) Maximum Profit / Weight
iii) Minimum Weight

UNIT V

1 Write the difference between BFS and DFS. (2 M)

2 Explain the Backtracking with example. (2 M)

3 Define Graph Colouring with example. (2 M)

4 Explain Promising and Non-Promising Node. (2 M)

5 Define tractable and non-tractable problems? (2 M)

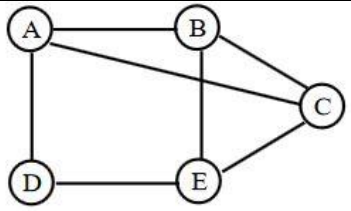
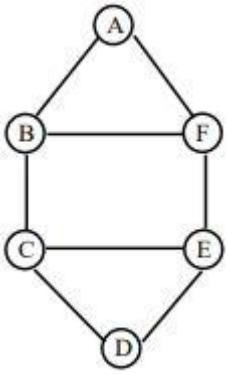
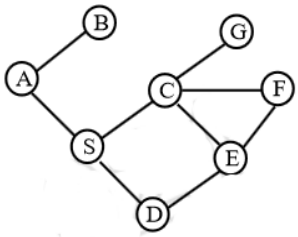
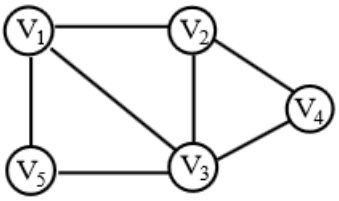
6 Discuss the classes P, NP, NP complete, and NP hard with examples. (4 M)

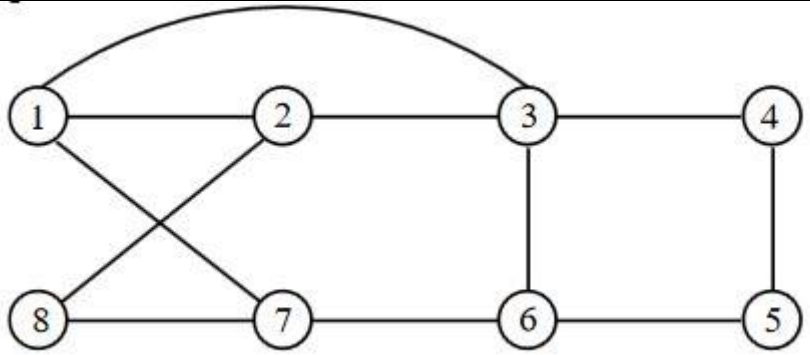
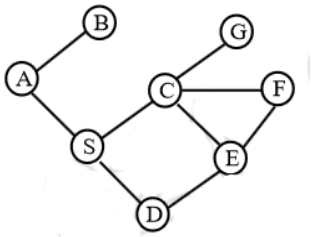
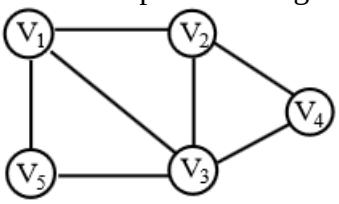
7 Give two solutions for 4-Queen Problem and 8-Queen Problem.(4 M)

8 State Cook's theorem and explain it. (4 M)

9 Write the algorithm for non - deterministic searching. (4 M)

10 Explain the graph colouring problem for the graph (undirected graph) given below. (7 M)

	
11	Justify the statement “Satisfiability is in P if and only if P = NP” (7 M)
12	Define Planner graph? Implement graph colouring on following graph and generate solution space tree. (7 M) 
13	Write an algorithm for following : (7 M) i) Clique ii) Graph Partitioned in Triangle. iii) Independent set problem
14	Write the algorithm for Breadth First Search (BFS). Also perform BFS on the given graph, showing operations using stack. State the output sequence generated.(7 M) 
15	Apply Graph Colouring on the given graph. State the total number of possible solutions by using solution space tree.(7 M) 
16	Write an algorithm for non-deterministic searching and non-deterministic sorting.(8 M)
17	Classify P, NP, NPC, NPH.(4 M)
	Define Hamiltonian cycle. Write the algorithm for the same & find the Hamiltonian cycle for the following graph. (8 M)

	
18	Explain following. (12 M) 1) NP hard & NP complete. 2) Characteristics of NP . 3) Non deterministic sorting. 4) Non deterministic searching.
19	Write Short notes on (12 M) i)P Class Problem ii) NP – Hard iii) NP – Complete iv) Deterministic Algorithm
20	Explain the different types of Graph traversal techniques.
21	Write the algorithm for Breadth First Search (BFS). Also perform BFS on the given graph, showing operations using stack. State the output sequence generated. (7 M) 
22	Perform Graph Colouring on the given graph. State the total number of possible solutions.(7 M) 
23	Differentiate between recursive and iterative algorithm design. (4 M)
24	State the difference between BFS and DFS.(2 M)
25	Write non-deterministic algorithm to generate CLIQUE of size k from the graph of n vertices.(5 M)
26	Explain decision and optimization problems?(4 M)
27	Explain following terms. (12 M) i) NP-Hard. ii) Polynomial Reduction.

	iii) Decision & optimisation problem.
28	Explain following NP-problems with respect to graph. (12 M) i) CLIQUE ii) Independent set problem iii) Graph partitioned into triangle.
29	Prove that P $\subseteq$ NP (5 M)

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